

URL SHORTENER

Functional requirements

1. Account creation -Register/Login
2. Convert long URL to short URL
3. History of converted URLs
4. Redirect to the shortened URL
5. Custom URL creation

Non -Functional requirements

1. Unique shortened URL
2. Available
3. Responsive

Estimations

- 10 million daily users
- Each user roughly shortens 2 URLs daily
- URLs shortened daily - 20 million
- size of 1 URL - 7 characters (alphanumeric values)
- size of 1 character - 1 bytes
- Total storage occupied for one day - 20 million * 7 * 1
=> 140 million bytes / 140 mb

Architecture

